

Supported Two-Phase Reading in Second Language Acquisition:

A Context-Based Approach to Promoting Receptive Competence

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STR project

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Abstract

Supported Two-Phase Reading (STR) combines secure initial comprehension with active retrieval of meaning in a compact two-stage process. In Phase 1, integrated translations and glosses ensure immediate understanding and maintain reading fluency. In Phase 2, the same passage is reread without any aids, prompting the learner to retrieve meanings from context and consolidate them as lexical items or expressions in the mental lexicon. The innovation of this approach lies in its refined, standardized protocol, which defines and measures a precise *fading moment* and includes verifiable outcome metrics (e.g., recall measures, L1-dependency index, cognitive-load indicators). STR thereby distinguishes itself from *interlinear-only* methods (support without obligatory retrieval), Birkenbihl-related approaches (explicit focus without a standardized retrieval phase), and parallel-text formats (parallel presentation without a binary phase shift).

Article Type

This paper is conceptual and positioning in nature; statements about effects are hypothetical and require empirical verification. The argument draws on recent meta-analyses on extensive reading, glossing, and retrieval practice (2010s–2020s). STR is presented as a standardized two-phase protocol: Phase 1 (P1) with interlinear support, and Phase 2 (P2) as a mandatory rereading without aids involving meaning retrieval. Testable hypotheses are derived (e.g., higher delayed recall, decreasing L1-dependency index), along with validation and adherence checks for the binary fading point. Since no original data are reported, all effect claims are provisional and require empirical replication in preregistered studies.

Originality and Contribution of the Study

Status of the Work

To the best of current knowledge, this is the first academic operationalization of a two-phase reading process with structured, binary fading. The study

1. systematically introduces the conceptual term into the scientific discourse,
2. establishes terminology and measurement tools (e.g., L1-dependency index, fading moment, autonomy score), and
3. sets methodological standards (preregisterable protocols, adherence/manipulation checks, and defined outcome measures).

Theoretical Contribution

– Bridges the gap between extensive reading and scaffolded instruction by linking comprehension

assurance (P1) with mandatory retrieval (P2) under binary fading (0 % support).

- Systematizes diverse practical approaches within a unified theoretical framework (from intensive to extensive reading; *transitional extensive reading*).
- Defines measurable constructs (e.g., L1-dependency index, fading moment) as a foundation for replication and meta-analytic synthesis.

Methodological Contribution

- Provides a standardized protocol for controlled studies on two-phase reading (time/exposure parity, preregistered analysis plans).
- Operationalizes *fading scaffolding* as a categorical phase shift (P1 → P2) with explicit adherence criteria.
- Standardizes outcome measures (e.g., delayed recall, load profiles, slope of the L1-dependency index), including process/layout audits and optional eye-tracking data.

Significance

The combination of a novel protocol structure, precise measurability, and open-science orientation enhances scientific comparability, replication, and evidence synthesis. It thus establishes a solid starting point for a research program on two-phase reading.

1. Introduction

The development of a stable receptive vocabulary and reliable text comprehension remains one of the central challenges in foreign language learning. Numerous reports and comparative studies indicate that learners often stagnate at elementary to lower-intermediate proficiency levels despite years of instruction. This discrepancy between effort and outcome points to structural weaknesses in conventional approaches, including excessive cognitive load, fragmented information processing, and interruptions to reading fluency.

Against this background, a context-based approach is proposed that places immediate comprehension at the center of learning while reducing dependence on isolated vocabulary and grammar exercises. The term *Supported Two-Phase Reading* (STR) refers to a procedure that combines comprehension assurance with autonomous meaning retrieval; the method is described in detail in Section 3. Its conceptual positioning, classroom observations, and learner feedback are summarized in Section 4.2 and in the appendix.

The objectives of this paper are threefold:

- (1) to outline the theoretical framework of STR within the context of relevant Second Language Acquisition (SLA) concepts,
- (2) to delineate its positioning in relation to comparable approaches, and
- (3) to outline a research design for empirical validation.

Nature of the study. This is a conceptual and positioning paper: we propose STR as a standardized two-phase protocol—Phase 1 involving interlinear support, and Phase 2 requiring mandatory meaning retrieval without assistance—and formulate testable hypotheses whose empirical validation remains to be established.

2. Theoretical Background

2.1 Theoretical Foundations

Comprehensible Input (Krashen, 1982, 1985)

The *i + 1* principle—input slightly above the learner’s current competence level—forms the basis of Phase 1 in STR. Translations and glosses close comprehension gaps, with the L1 serving as a temporary

crutch to ensure immediate intelligibility. Similar to how children acquire language through comprehensible contexts, adults are offered a low-stress entry point to meaningful reading.

As Long (1996) emphasizes, comprehensible input alone is insufficient; effective learning also requires interaction and negotiation of meaning. STR responds to this by systematically coupling comprehensible input (Phase 1) with mandatory retrieval (Phase 2), thus extending beyond a purely input-based perspective.

Reducing Cognitive Load (Sweller, 1988; Bonilla Carvajal, 2025)

Bonilla Carvajal (2025) demonstrated in an experiment with 48 EFL beginners that interlinear translations not only reduced cognitive load but also significantly increased the number of correctly translated words in both immediate and delayed tests. While this finding confirms the benefits of interlinear translation for Phase 1, Phase 2 of STR extends it through the systematic operationalization of *retrieval practice* (Nation, 2001). This supports durable lexical integration and the transition toward linguistic autonomy.

These findings should be considered preliminary (n = 48, not yet replicated); a preregistered replication is planned within the STR framework.

Cognitive Load Theory (Paas et al., 2003; Sweller et al., 2019)

Research demonstrates how effective instructional design can minimize *extraneous load* and strengthen *germane load*. STR operationalizes precisely this mechanism: Phase 1 reduces processing burden, whereas Phase 2 deliberately induces cognitively demanding retrieval.

Scaffolding and the Zone of Proximal Development (Vygotsky, 1978)

In Phase 1, learners receive clearly structured support through green-highlighted translations and glosses, ensuring immediate comprehension. In Phase 2, this support is completely withdrawn, requiring independent understanding based on the foundation established in Phase 1. This approach represents an operationalization of *fading scaffolding*: as linguistic competence grows, assistance becomes progressively less relevant until it is no longer needed.

STR thus concretely implements Vygotsky's concept of the *Zone of Proximal Development*: support is available as long as necessary and systematically reduced once independent comprehension becomes possible. Comparable fading mechanisms are known from strategy-training systems such as *iSTART*, where a standardized reduction of support proved beneficial for learning (McNamara et al., 2006).

Motivation and Autonomy

A central mechanism of STR lies in its motivational effects. In Phase 1, learners experience immediate success: the text is comprehensible, frustration is avoided, and reading enjoyment increases. This reduces anxiety and enhances willingness to continue reading—contrasting with traditional methods, where incomprehension often leads to demotivation (cf. Bandura, 1997; Schunk & Pajares, 2002). Phase 2 further strengthens the perception of competence: learners realize they can retrieve meanings without assistance. This experience fosters *self-efficacy* and provides a foundation for developing reading autonomy.

These effects align with *Self-Determination Theory* (Ryan & Deci, 2020): Phase 1 supports autonomy and perceived competence through immediate comprehensibility, while Phase 2 reinforces perceived competence through successful retrieval without aids.

Depth of Processing (Craik & Lockhart, 1972)

Rereading the same passage promotes semantic integration. During the repetition phase, *retrieval practice* is activated, which facilitates long-term retention. Neurolinguistic evidence (Shen et al., 2021) indicates that context-bound vocabulary learning elicits stronger neural activation patterns that promote durable consolidation of lexical items in the mental lexicon. Findings from affective-memory research further suggest that contextualized learning enhances emotional memory effects (Cahill & McGaugh, 1998; Kensinger & Corkin, 2003), potentially strengthening retention. In STR, affective influences are

addressed primarily through contextual processing and retrieval; emotion-related findings outside L2 settings are discussed elsewhere.

Principles of Repetition (Nation, 2001)

Through the cognitive relief provided in Phase 1, learners can read larger quantities of text in which the same vocabulary frequently recurs. Such repeated encounters with words in natural context align with Nation's principle of *meaning-focused repetition*. Phase 2 amplifies this effect by requiring re-processing of words without support. Thus, STR combines low cognitive load with high exposure density, fostering sustainable vocabulary acquisition.

Empirical Evidence (Krashen, Long, Nation, Horst, Waring)

Empirical research supports these assumptions. Krashen and Long emphasize the centrality of comprehensible input for language acquisition, while Nation (2001) highlights the effectiveness of repeated, context-based reading. Craik and Lockhart (1972) demonstrated that contextual depth of processing enhances information retention. Horst (2005) and Waring (2009) likewise found that extensive reading can produce substantial vocabulary gains—up to 500–1000 new words per month in some studies. Collectively, these results suggest that contextual, repeated reading is both theoretically plausible and empirically supported, even though some studies (e.g., Taylor 2014) report limited or nonsignificant effects of L1 glosses.

Glossing: Evidence, Moderators, and Implications for STR

Evidence (1997–2025). Early studies found that interlinear embedding improves text comprehension and short-term vocabulary gains (Watanabe, 1997). Eye-movement data indicate that interlinear glosses reduce *split-attention effects* (shorter saccades and regression rates), thereby supporting reading fluency (Warren et al., 2018). Meta-analyses consolidate this advantage: interlinear glossing is consistently more effective than marginal or end-placed glosses (Yanagisawa, Webb & Uchihara, 2020); multimodal glosses (text + audio/image) yield additional benefits (Ramezanali, Uchihara & Faez, 2021; Mahdi et al., 2024). For lower-proficiency learners, L1 glosses often outperform L2 glosses by reducing processing effort and ensuring immediate access to meaning. A meta-analysis reports robust overall effects of glossing and identifies moderators related to format and learner characteristics (Yanagisawa et al., 2020).

Constraining evidence and moderators. Results remain heterogeneous: in *Taylor-type* studies, effects of L1 glosses are sometimes neutral or limited; in poorly designed tasks, glossing may even induce overload or superficial processing (Cook, 2001; Taylor, 2014). Variability can be explained by moderators such as placement (interlinear vs. marginal/end), gloss density and quality, proficiency level, text difficulty, and testing conditions (recognition vs. recall, immediate vs. delayed). A recent review (Boers, 2022) calls for cautious interpretation of meta-analytic data, emphasizing ecological validity and implementation details. New experimental data with beginners further indicate that interlinear translation reduces cognitive load and improves delayed retention (Bonilla Carvajal, 2025).

Implications for STR (Design Response).

Phase 1 integrates aids interlinearly within the reading flow—minimizing split attention and extraneous load—and thereby leverages the empirically validated strengths of glossing. The innovation of STR lies in the mandatory Phase 2: all aids are categorically removed (*fading*), and meaning retrieval becomes obligatory (*retrieval practice*). This design systematically mitigates the risk of L1 overreliance discussed in prior research: L1 support is confined to Phase 1, whereas Phase 2 enforces autonomous recall in the target language. Clear empirical predictions follow: higher delayed-recall rates, a decreasing L1-dependency index across sessions, and distinct eye-tracking/pupillometry patterns between Phase 1 (relief) and Phase 2 (targeted activation).

Empirical confirmation from related research.

The effectiveness of the two-phase structure finds support across multiple domains:

– **Glossing and translation in reading:** Elekaei et al. (2015) showed that interlinear glosses

significantly improved comprehension and vocabulary retention compared with marginal glosses or glossaries—supporting Phase 1 as an integrated scaffold.

– **Repeated reading and assisted vocabulary learning:** Soleimani et al. (2022) found that repeated reading with auditory support produced higher vocabulary gains than unassisted reading—confirming the “assisted repeated reading” principle operationalized in STR through translation + repetition.

– **Testing effect and active retrieval:** Kang et al. (2013) demonstrated that active meaning retrieval yields longer-term vocabulary gains than passive repetition without impairing pronunciation—paralleling the logic of Phase 2.

– **Extensive Reading:** Meta-analyses (e.g., Nakanishi, 2015) and reviews (Horst, 2005; Nation, 2015) confirm positive effects of extensive reading on vocabulary and motivation. STR can thus be viewed as a *bridging approach* (IR → ER), leveraging the strengths of extensive reading through a standardized two-phase structure (integrated support → categorical fading with mandatory retrieval).

– **Scaffolding and ZPD:** Beyond Vygotsky (1978), the seminal study by Wood, Bruner & Ross (1976) underscored that temporary support, systematically reduced over time, is crucial for autonomous competence development. STR applies this principle directly: support in Phase 1, fading in Phase 2.

2.2 Didactic Context

The approach closely parallels *Extensive Reading* (ER), an established pedagogical strategy in which linguistic competence develops through meaning-oriented reading of large quantities of text, focusing on comprehension rather than dictionary use. Numerous international studies (e.g., Nation, Long, Richards) demonstrate improvements in vocabulary, grammatical intuition, and reading fluency, though effects vary by context.

STR is conceptualized not as a subtype of ER but as a *bridging approach* between intensive and extensive reading (*transitional extensive reading*). It remains ER-compatible, as it employs narrative long-form texts (e.g., adapted classics at A1–A2 or originals at B1–B2 levels) that realize ER’s hallmark of meaning-focused reading across large text volumes.

Current evidence. Meta-analyses report small-to-moderate effects of ER on reading comprehension, fluency, and vocabulary (Jeon & Day, 2016; Nakanishi, 2015). Recent syntheses on meaning-focused input quantify incidental vocabulary gains and identify key moderators (Webb et al., 2023); updated overviews summarize both research trends and pedagogical applications (Zhou, 2024).

2.3 Potential of AI-Supported Text Adaptation

A custom-configured GPT or similar AI-based tool can automatically simplify any authentic text and convert it into the two-phase STR format. First, the text can be simplified (at the learner’s request), after which step-by-step translations and short glossaries are integrated (Phase 1). The same passage is then presented again—without support (Phase 2).

This enables highly personalized reading pathways:

- Beginners can automatically simplify authentic texts, made immediately readable through integrated translations and glosses in Phase 1.
- Advanced learners can process more complex materials with graded support.
- Teachers can create differentiated and adaptive learning materials.

Moreover, AI systems can dynamically adjust translations, glosses, and repetition intervals to match individual proficiency levels. Recent studies suggest that AI-based learning environments offer significant benefits for vocabulary acquisition and self-regulated learning (Huang et al., 2023; Li et al.,

2024; Liu & Chiu, 2023; Müller & Schneider, 2025; Rahman, 2025). These findings highlight the relevance of STR, whose clear two-phase structure can be readily integrated into AI-supported training formats.

To operationalize this concept, we developed an open-source *STR Annotation System* (v2.0), a reproducible AI-based pipeline for generating two-phase reading materials in **HTML format**. The system integrates GPT-class models with a strict quality-control protocol (literal accuracy, HTML validation, and P2–P1 identity checks) and automatically produces dual-phase HTML texts ready for research or publication.

Each output undergoes validation against the STR quality checklist, ensuring full compliance with the standardized two-phase protocol (P1 support → P2 retrieval).

This implementation demonstrates that AI-supported text adaptation can meet scientific standards of reproducibility and comparability while significantly reducing human workload in material preparation.

At the same time, automatic AI translations are not always fully accurate. Teachers therefore play a crucial role in reviewing and refining texts for precision and naturalness. Publishers that release books following the STR method typically conduct such editorial revisions as a standard procedure—similar to any conventional publication process.

3. Methods

3.1 Overview

Supported Two-Phase Reading (STR) is a context-oriented approach that realizes *fading scaffolding* as a binary, passage-bound shift: in Phase 1, translations and glosses are provided in full and interlinearly; with the hard transition to Phase 2, all aids are removed so that the same passage is read without support and meanings are actively retrieved. In this way, the Zone of Proximal Development (ZPD) is operationalized pedagogically: temporary, closely guided support followed by independent performance.

Terminology (consistent use). Throughout this paper, we use the term *fading scaffolding* in the specific sense of STR; the first mention is italicized, subsequent mentions are not.

Definition (fading scaffolding in the sense of STR). In this study, *fading scaffolding* denotes a phase-based, categorical removal of all linguistic aids (translations/glosses) after an initial support phase. In contrast to continuous fading approaches—where supports are reduced gradually and adaptively (e.g., lower density, selective glossing)—fading in STR is binary and passage-bound: Phase 1 → Phase 2.

Operationalization in STR.

1. Phase 1: full interlinear support embedded in the reading flow (translations/glosses for all relevant units).
2. Hard phase switch with no intermediate steps or residual aids.
3. Phase 2: the identical passage presented completely without translations/glosses (mandatory meaning retrieval).
4. Pedagogical consequence: split attention is minimized in Phase 1; autonomy is strengthened in Phase 2 through retrieval practice.

Measurable criteria (for studies). Proportion of aids by phase (P1 $\approx$ 100%, P2 = 0%), a hard phase indicator (binary switch), recall measures in P2 (immediate/delayed), and subjective/objective load indicators.

Relation to the ZPD. The two-phase scheme is a pedagogical operationalization of the ZPD: temporary, closely guided support in P1; targeted removal of aids and independent performance in P2. STR thus clearly differs from approaches that merely “thin out” supports without removing them completely.

The method is described conceptually in what follows; controlled studies of effectiveness are still pending. Initial qualitative evidence comes from user feedback (see reviews and comments), which indicates practical applicability. We define *fading scaffolding* as a binary, paragraph-bound transition from P1 (with aids) to P2 (without aids). The fading point is specified *a priori*; P2 contains 0% aids. Adherence/manipulation checks include (a) a layout audit (confirmation of 0% aids in P2), (b) process markers (e.g., eye-tracking evidence of minimal gloss use in P1 and absence in P2), and (c) system logs. Terminology is standardized (e.g., “language autonomy”); redundant definitions are omitted.

3.2 Procedure

STR implements *fading scaffolding* as a binary, passage-bound shift: Phase 1 (full support) → Phase 2 (complete removal of aids).

Phase 1 — Reading with support.

The text appears in the original, with translations and brief glosses integrated in green within the reading flow:

Note: The following illustrates Phase 1 with L1 (German) interlinear glosses for demonstration purposes.

When Mr. Hiram B. Otis, the American Minister (als Mr. Hiram B. Otis, der amerikanische Gesandte), bought Canterville Chase, everyone told him (Schloss Canterville kaufte, warnten ihn alle; *to buy, bought, bought – kaufen; chase – das Jagdschloss, das Landgut; hier: Schloss Canterville*) it was a bad idea (es wäre eine schlechte Idee). The place had a ghost (dort soll es gespuht haben). There was no doubt about that (daran bestand kein Zweifel; *doubt – der Zweifel*). Lord Canterville was a man (Lord Canterville war ein Mann) who paid attention to detail (der großen Wert auf Details legte; *to pay attention to smth. – Aufmerksamkeit schenken, aufpassen, achten auf, sich kümmern um; to pay – bezahlen; attention – die Aufmerksamkeit, die Achtung, die Beachtung*). He felt it was his duty (er sah es als seine Pflicht an; *to feel, felt, felt – fühlen, sich fühlen, sich befinden, meinen, spüren, glauben*) to tell Mr. Otis about the ghost (Mr. Otis über den Geist zu informieren; *to tell, told, told – sagen, erzählen, berichten*). This happened (dies ereignete sich; *to happen – passieren, geschehen*) when they sat down (als sie sich setzten; *to sit down – sich setzen; to sit, sat, sat – sitzen*) to discuss the terms (um über die Kaufbedingungen zu sprechen; *to discuss – diskutieren, besprechen, erörtern; terms – Bedingungen, Konditionen*).

Color-coding enables learners to consult translations instantly when needed—or to skip them effortlessly when the meaning is already known. This creates a smooth transition from supported to independent reading.

Phase 2 — Reading without support.

Immediately afterward, the same passage is presented again without translations or comments. Meanings must then be actively retrieved from memory (retrieval practice), which supports long-term consolidation in the mental lexicon and reduces reliance on the L1.

When Mr. Hiram B. Otis, the American Minister, bought Canterville Chase, everyone told him it was a bad idea. The place had a ghost. There was no doubt about that. Lord Canterville was a man who paid attention to detail. He felt it was his duty to tell Mr. Otis about the ghost. This happened when they sat down to discuss the terms.

This example illustrates the core methodological principle: contextual meaning construction (Phase 1) followed by immediate repetition without aids (Phase 2).

Flexible application.

Typically, beginners start with Phase 1. As reading experience grows, the need for support decreases, making it easy to skip color-highlighted translations for already familiar items. Advanced learners may optionally invert the sequence: start directly with Phase 2 and consult Phase 1 only as a supplementary reference. This is not the standard procedure, but an adaptation for higher proficiency levels. In this way, STR can be flexibly aligned with different proficiency levels without altering the underlying principle.

The cognitive relief provided by Phase 1 is empirically supported (see Section 2.1).

Retrieval & spacing. An L2 meta-analysis shows that spaced practice outperforms massed practice—especially on delayed tests (Kim & Webb, 2022). General meta-analyses on spaced retrieval corroborate this logic (Latimier et al., 2021). Recent experiments also indicate that covert and overt retrieval can be similarly effective for L2 vocabulary, allowing flexible implementation of Phase 2.

3.3 Materials and Text Selection

Text selection follows Nation's (2001) criteria, according to which beginners benefit most from a focus on high-frequency lexicon (approx. 2,000–3,000 word families). Work with narrative long-form texts is consistently prioritized, as extended narrative increases exposure, supports immersion and repetition of core items, strengthens reading fluency, and is therefore compatible with the principles of Extensive Reading.

At A1–A2 levels, pedagogically adapted classical and contemporary texts are used in level-appropriate versions. The high repetition density of high-frequency vocabulary within longer narratives facilitates meaning construction and supports gradual automatization.

At B1–B2 levels, unmodified literary originals (long form) are employed, with translations and brief glosses integrated directly into the reading flow. Learners thus work with authentic language without layout changes or interruptions to fluency; at the same time, long-form reading is preserved, maximizing meaning-focused (extensive) processing across chapters.

A defining feature of the approach is the use of longer, narratively structured texts. Unlike isolated exercises or brief fragments, coherent stories promote sustained engagement with language. They increase motivation through narrative tension, enable immersion in the flow of language, and offer frequent opportunities for context-based repetition of key vocabulary and structures. Continuous narration also reduces cognitive load, as meanings can be inferred from the ongoing context. Finally, long-form work supports self-directed learning, allowing learners to experience progress within the text and to adjust their pace individually.

Empirical findings support these assumptions: Extensive Reading studies (Nation, 2001; Waring, 2009; Horst, 2005) show that longer, narratively structured texts increase motivation and immersion, accelerate vocabulary growth, and reduce cognitive load. Cognitive Load Theory (Sweller, 1988; Paas et al., 2003) likewise indicates that continuous narratives reduce extraneous load, as meanings can be inferred from context. Newer neurolinguistic studies (Shen, Li & Sun, 2021) further confirm that context-based vocabulary learning elicits more stable neural activation patterns than isolated practice, facilitating durable integration of new words.

Formats. All materials are available as e-books, print editions, and audio recordings by native speakers. Each text is accompanied by an audio version so that reading and listening can be combined, additionally supporting listening comprehension, intonation, and pronunciation.

Design. Texts are divided into short sections, aids are highlighted in color, and the layout is optimized for clarity and readability—including on digital devices.

3.4 Methodological Limitations

- **Focus on receptive skills.** The approach primarily strengthens reading comprehension and vocabulary; productive skills (speaking, writing) are only indirectly affected.
- **Dependence on translation.** Exclusive use of Phase 1 risks an L1 “crutch.” STR addresses this risk through the mandatory Phase 2, but the transition requires careful management.
- **Text selection.** Effectiveness depends heavily on the fit between materials and proficiency level. Texts that are too complex can be demotivating.

4. Empirical Support

4.1 Evidence from Research

While the underlying techniques have long been known, the distinctive contribution of STR lies in the protocol-level standardization of their binary coupling: Phase 1 → hard switch → Phase 2, with a defined fading point and operationalized outcome measures. Related approaches—such as interlinear reading followed by reading the original, commercial series with sublines (literal parallel translations), or Birkenbihl-adjacent methods—have existed for some time. What is novel in STR is not any single technique, but the standardized, binary coupling: Phase 1 (full interlinear support) → Phase 2 (complete removal of aids) with mandatory meaning retrieval (retrieval practice). This yields a testable protocol that, for the first time, makes the integrative value of these familiar elements systematically measurable. For cognitive relief, see Section 2.1.

Table 1. Empirical Evidence for Core Principles of Supported Two-Phase Reading (STR)

Principle	Empirical Evidence	Relevance to STR
Comprehensible Input	Krashen (1982, 1985); Long (1983); Liu & Chiu (2023): Input slightly above the learner’s current proficiency level facilitates acquisition.	Phase 1 closes comprehension gaps through translations and glosses, making texts accessible at lower levels (A1–B2).
Repetition / Retrieval	Nation (2001); Kim & Webb (2022); Li et al. (2024): Repeated exposure and spaced practice improve long-term retention, especially on delayed tests.	The direct transition from Phase 1 to Phase 2 elicits immediate meaning retrieval (<i>retrieval practice</i>).
Context-Based Reading	Horst (2005); Waring (2009); also emotional-memory effects: Cahill & McGaugh (1998); Kensinger & Corkin (2003).	STR uses narrative texts whose thematic tension promotes depth of processing.
In-Flow Support (Interlinear Embedding)	See overview and references in Section 2.1.	Phase 1 integrates aids directly into the reading flow; Phase 2 enforces autonomous retrieval.
Depth of Processing	Craik & Lockhart (1972); Shen et al. (2021); Perdue (2021); Braid & Richlan (2022): Context-bound learning correlates with more stable neural activation.	Rereading the same passage in Phase 2 promotes integration of new lexis in context.

Principle	Empirical Evidence	Relevance to STR
Motivation / Self-Regulation	Dörnyei & Ryan (2015); Huang et al. (2023): Adaptive, interactive environments enhance motivation and autonomy.	Initial practice reports indicate similar effects: immediate comprehensibility increases motivation and self-direction.

Relevance to STR: Initial practice reports suggest analogous effects: immediate comprehensibility boosts motivation and self-direction.

Note. An integrated treatment of glossing evidence (including critical findings and the design response) is provided in Section 2.1; below we refer only to operational implications for the research program.

4.2 Practice Observations

In addition to theoretical considerations, there is a substantial body of anecdotal learner feedback compiled from reviews and email correspondence. These reports point to recurring motivational, cognitive, and affective effects that align with the mechanisms summarized in Table 1. A detailed documentation of these experiences is provided in the appendix.

5. Discussion

5.1 Comparison with Existing Approaches

As outlined in Section 2.1, interlinear glossing functions primarily as a tool for ensuring comprehension. STR extends this principle through a mandatory retrieval phase (see Section 3) and is conceptually positioned as a *bridging approach* (IR → ER)—a form of *transitional extensive reading* that combines intensive meaning construction in Phase 1 with extensive, aid-free reading in Phase 2. Because it employs long-form texts, the method remains compatible with Extensive Reading (see Section 2.2). The following comparison focuses on core criteria and deliberately avoids repeating descriptions of the two phases or previously summarized glossing evidence.

Table 2. Comparison of Reading Strategies by Core Characteristics

Approach	Comprehension Support	Retrieval Autonomy	Cognitive Load	Reading Flow / Split-Attention	Risk of L1/L2 Crutch	Positioning
Extensive Reading	Low (level-dependent)	Medium (indirect, long-term)	High for beginners	Seamless / low	Low	Core Extensive Reading framework
Parallel Texts	High (direct comparison)	Medium	Medium–high (layout switching)	Interrupted / high	Medium	Parallel-text approach
Glossing (marginal/end)	Localized	Low	Medium	Interrupted / moderate	Low	Glossing without fading
Interlinear Reading	Very high	Low	Very low	Seamless / low	High	Interlinear, without fading

Approach	Comprehension Support	Retrieval Autonomy	Cognitive Load	Reading Flow / Split-Attention	Risk of L1/L2 Crutch	Positioning
STR (two-phase)	High	High (mandatory)	Low–medium	Seamless / low	Low (through fading)	Bridging approach IR → ER (<i>transitional extensive reading</i> , ER-compatible)

Note: For empirical evidence on interlinear embedding, see Section 2.1; for methodological details on STR, see Section 3.

STR vs. Repeated Reading (Meta-analysis: Choi & Zhang, 2022)

Meta-analyses on *Repeated Reading (RR)* show robust but moderate effects on reading fluency and, to some extent, on vocabulary and comprehension. Heterogeneity is moderated by text difficulty, test format (immediate vs. delayed), and learner proficiency. RR does not systematically ensure comprehension and lacks a defined retrieval phase without aids.

Positioning of STR (refined protocol).

STR can be described as “RR with controlled scaffolding and enforced retrieval.” Phase 1 minimizes *extraneous load* through integrated support; the measurable *fading moment* then introduces a binary transition to Phase 2, where active retrieval without aids is mandatory. STR thereby operationalizes precisely those components that often vary in RR designs (amount of support, timing, and retrieval requirements). It directly addresses the moderator variables identified in Choi & Zhang’s (2022) meta-analysis as causes of heterogeneity (text difficulty, testing window, learner level). Through measurable operationalization of the fading point, STR allows direct and replicable comparisons.

Fair comparison logic (design rationale).

Direct comparisons between STR and RR assume *time/exposure parity* (identical text amounts, repetitions, and testing intervals). Manipulation checks for fading (layout audit, eye-tracking) ensure that Phase 2 is indeed aid-free.

Expected data patterns (preregisterable).

- Initial comprehension (global/local): $\text{STR} \geq \text{RR}$ due to guaranteed initial understanding in Phase 1.
- Delayed recall/retention: $\text{STR} > \text{RR}$ with equal exposure times (retrieval advantage from Phase 2).
- L1 dependency index: steeper negative slope across sessions under STR.
- Reading fluency: short-term parity or slight disadvantage in Phase 2 (retrieval cost), converging mid-term.
- Load profiles: Phase 1 (STR) $\leq$ RR in extraneous load; Phase 2 shows targeted activation (longer response times/pupil dilation) correlated with better delayed outcomes.
- Motivation/attrition: lower dropout rates under STR due to immediate comprehension in Phase 1.

Relation to the meta-analysis.

The variance reported by Choi & Zhang (2022) is incorporated within the STR framework as a spectrum of moderators (proficiency, difficulty, test format). The unique contribution of STR lies in its measurable fading moment and protocol fidelity (adherence checks), which allow more precise and replicable estimation of effects (see Chapter 7 for operationalization and preregistration logic).

Beyond RR, further comparisons address structural and functional contrasts with other common reading aids.

STR vs. Interlinear-Only

Interlinear-only reading ensures comprehension but does not enforce retrieval without aids. STR introduces a mandatory second phase with complete fading.

Expected data: higher delayed recall rates (retrieval tests) under STR; lower L1-dependency index across sessions; equal or lower extraneous load in Phase 1 (eye-tracking: shorter fixations) but targeted activation in Phase 2 (longer processing time, pupil dilation). Metrics and indices are defined in the research program.

STR vs. Marginal Glosses

Marginal glosses interrupt the reading flow (layout switching) and dilute support. STR integrates aids interlinearly (Phase 1) and removes them categorically (Phase 2).

Expected data: better immediate comprehension (Phase 1) and higher delayed retention (Phase 2) under STR; lower split-attention markers (eye-tracking regression rates) in Phase 1.

Meta-analytic findings on glosses: multimodal and line-proximal glosses reliably facilitate L2 vocabulary learning across tests (Ramezanali, Uchihara & Faez, 2021). Meta-analyses confirm consistent overall effects and moderators by format, language, and mode (Yanagisawa, Webb & Uchihara, 2020). For lower proficiency levels, L1 glosses are often advantageous; multimodal glosses yield additional gains (Mahdi, 2024). These findings support Phase 1 of STR (integrated, layout-stable) and justify complete fading in Phase 2.

STR vs. Parallel Texts

Parallel texts provide permanent side-by-side support but lack a defined retrieval phase. Moreover, literary translations are rarely word-for-word equivalents: to achieve idiomatic naturalness and target-language fluency, translators employ transposition, modulation, omission, or addition. This microstructural non-isomorphism undermines the one-to-one mapping of form and meaning at the word/phrase level—learners cannot reliably extract intended semantics even when lines appear parallel. Critical studies on glossing also highlight risks of L1 overreliance and inconsistent form–meaning mapping, particularly in literary translation, which can blur intended meanings (Taylor, 2014; Boers, Warren, Grimshaw & Siyanova-Chanturia, 2017; Boers, 2022).

Positioning (design response).

STR avoids layout switching, integrates aids interlinearly, and compensates for missing word-for-word equivalence through expanded comments and annotations (lemma/base form, multiple meanings, valency, morphosyntactic cues, collocations, pragmatic nuances). Where no direct correspondence exists, paraphrasing, function labels, and comparative examples provide explicit meaning bridges. The standardized fading moment then removes all aids (Phase 2), normalizing retrieval. Through these features, STR directly addresses issues discussed in the glossing literature and limits L1-bias effects (Taylor, 2014; Boers et al., 2017).

Expected data: under STR—higher reading fluency in Phase 1 (words/min; lower split-attention and regression rates) and higher recall in Phase 2 (especially delayed). For parallel texts—higher split attention, persistent L1 use, and less precise form–meaning mapping.

STR vs. Peer-Reviewed Baselines

Commercial two-phase reading formats (e.g., interlinear + repetition, such as the *Ilya Frank series*) exist in popular language-learning materials but lack standardized protocols for research purposes. **Therefore**, for academic comparison, STR is aligned only with peer-reviewed baselines that share structural similarities but differ in operationalization.

Peer-reviewed baselines:

- **Assisted Repeated Reading (ARR; Soleimani et al., 2022)** – repetition and audio support, but no binary fading or adherence control.
- **Interlinear glossing (Watanabe, 1997; Warren et al., 2018)** – integrated support ensures comprehension, but retrieval without aids is not required.
- **Continuous/adaptive fading (McNamara et al., 2011, iSTART)** – gradual reduction of support; STR employs categorical removal (0 % support in P2).

STR innovation:

STR introduces a preregistered binary fading moment, measurable adherence checks (layout audit, eye-tracking), and new outcome metrics (L1-dependency index, autonomy score). This protocolization transforms a pedagogical sequence (support → retrieval) into a testable research framework.

Expected outcomes:

STR is expected to yield steeper declines in L1 reliance and higher delayed recall compared to ARR and interlinear-only conditions under equal exposure time.

Synthesis

Taken together, meta-analyses suggest that integrated support (Phase 1) combined with mandatory spaced retrieval (Phase 2) optimizes the conditions under which Extensive Reading, glossing, and retrieval practice exert their well-documented effects.

Neither interlinear translation nor repeated reading are new in themselves; the innovation of STR lies in the standardized binary coupling with a defined fading moment, preregisterable outcome measures (e.g., L1-dependency index, delayed recall), and adherence checks. STR is therefore a *protocol*, not merely a *technique*. Overlaps with existing approaches are explicitly acknowledged.

Interpretation and Positioning.

Table 2 shows that STR combines strong comprehension assurance with high autonomous retrieval while maintaining moderate cognitive load; risks of sustained L1 dependence are mitigated through categorical fading in Phase 2 (see Section 3). STR thus positions itself as a bridging approach (IR → ER) that unites integrated support (Phase 1) with obligatory, aid-free retrieval (Phase 2). It is distinct in its two-phase logic and clearly differentiated from interlinear reading (support without retrieval), marginal glossing/parallel texts (permanent layout-dependent support without defined retrieval moment), and repeated reading (retrieval without integrated support). Whether the expected advantages are confirmed in direct comparative studies remains an open question for future empirical validation (see Chapter 7).

Critical Appraisal

The present argument acknowledges both the strengths and the limitations of glossing and scaffolding-based reading approaches. As Peters et al. (2019) demonstrate, excessive use of L1 glosses can lead to shallow processing—STR addresses this through binary fading. The logic of the model also aligns with McNamara et al. (2011): in *iSTART*, it was shown that binary scaffolding/fading can enhance training effects; STR explicitly transfers this principle to reading by having Phase 1 (full support) culminate in a categorical shift to Phase 2 (0% support, mandatory retrieval). Moreover, Open Science recommendations (Marsden et al., 2019) place preregistration at the core of good research practice. STR is particularly well suited to this, as the *fading moment* represents a clearly preregistrable event. These points frame the hypotheses regarding delayed recall, decreasing L1 dependency, and differentiated load profiles (P1 relief vs. P2 targeted activation), providing a transparent foundation for replicable tests in controlled designs.

Monolingual Frameworks and Bilingual Memory

The theoretical argument builds partly on the *Levels of Processing* model (Craik & Lockhart, 1972). However, this framework was developed primarily in monolingual contexts. Research on bilingual memory demonstrates that emotional coupling, encoding, and retrieval can vary systematically between L1 and L2 (Ayçiçeği-Dinn & Caldwell-Harris, 2009). For STR, this implies potential moderation effects related to language pair, acquisition history, and degree of bilingualism: the strength of Phase 1 effects (support) and retrieval quality in Phase 2 may vary across language configurations. Findings from bilingualism research also show that interference and language mode moderate L2 processing (Grosjean, 2010; see also Kroll & Dussias, 2017); such profiles are accounted for in our analyses.

Usage-Based Perspective

From a *usage-based/emergentist* perspective (Ellis, 2008), language development results from frequency-based consolidation of form–function mappings and constructions. STR effectively targets both frequency and semantic clarity (Phase 1) and supplements these with *retrieval practice* (Phase 2). Nevertheless, the protocol primarily emphasizes lexical-semantic integration; systematic construction-level generalization remains peripheral. A potential enhancement could involve brief, construction-focused micro-retrievals in Phase 2 (e.g., sentence-pattern-based retellings) to strengthen explicit form–function mappings.

Input vs. Interaction

STR's framework draws on the concept of *comprehensible input* (Krashen, 1982, 1985) and endorses Long's (1983) emphasis on input slightly beyond current competence. However, Long also stresses the role of *negotiation of meaning* and interactive processes, while critiques of the Input Hypothesis argue that input alone is insufficient. STR responds to this by systematically coupling input with retrieval; what remains open is the extent to which interactive or output components (e.g., brief oral summarization after Phase 2) should be incorporated. As Long (1996) notes, comprehensible input alone is not enough for durable L2 acquisition—interactive negotiation is also essential. STR acknowledges this by combining input with obligatory retrieval, thereby extending beyond the classical Input Hypothesis.

Heterogeneous Evidence on Glossing

Meta-analyses report robust advantages of glossing, especially when placed near sentences or lines and implemented multimodally (Yanagisawa, Webb & Uchihara, 2020). However, other syntheses and primary studies report limited or neutral effects—particularly for L1 glosses and poorly designed formats (Taylor, 2014). STR addresses this heterogeneity through interlinear embedding in Phase 1 and

categorical fading in Phase 2. Whether this compensates for the limitations described by Taylor (2014)—such as shallow processing or L1 overreliance—remains an empirical question. Peters et al. (2019) also highlight the risks of excessive L1 glossing, which may lead to superficial processing and delayed L2 autonomy. These critiques underscore that both the design and dosage of support are critical. Boers (2022), summarizing the state of the art, confirms this mixed evidence on glossing and multiword units, cautioning against overreliance on L1 aids in suboptimal designs.

Open Questions and Testable Predictions (Link to Chapter 7)

- (i) **Bilingualism moderation:** Interaction effects of *language pair* $\times$ *phase* on delayed recall and L1-dependency index.
- (ii) **Constructional learning:** Additional measures of form–function mappings (beyond lexical tests) to test usage-based predictions.
- (iii) **Mechanism triangulation:** Eye-tracking/pupillometry to contrast relieved processing (Phase 1) vs. targeted activation (Phase 2); expected: lower regression rates in Phase 1, stronger retrieval markers in Phase 2.
- (iv) **Glossing moderators:** Systematic variation of placement/density/modality of support to reconcile the divergent findings of Yanagisawa et al. (2020) vs. Taylor (2014).

This critical appraisal positions STR as compatible with input-based, usage-based, and interactionist perspectives, while emphasizing that its hypothesis-driven effects—particularly the reduction of L1 dependence through *fading scaffolding*—require direct empirical verification under controlled conditions.

Scope (Romance and Slavic Languages)

The present framework intentionally focuses on Romance and Slavic languages. Even within this scope, contexts differ in terms of case and agreement marking, aspect (PFV/IMPF), clitics, and word order variation. Accordingly, glosses in Phase 1 include minimal morphosyntactic annotations (case/aspect/clitic/verb inflection) that are systematically reduced across chapters (*fading*). Productive transfer after Phase 2 (oral recall, dictogloss, short writing tasks) is assessed, and L1 dependence is measured longitudinally. This design addresses the heterogeneity of glossing effects (Yanagisawa 2020 vs. Taylor 2014) without overgeneralizing to typologically distant systems (e.g., agglutinative languages).

5.2 Limitations and Risks

Despite the advantages outlined, STR's effects are bounded. The procedure primarily strengthens receptive skills; productive skills such as speaking or writing benefit only indirectly. Learning outcomes also depend heavily on self-motivation and sustained engagement—an advantage and a risk in self-study contexts.

A further critical factor is the quality and suitability of text selection. Inappropriate or overly complex materials can impede comprehension, increase cognitive load, and reduce motivation.

Particularly noteworthy is the risk of translation dependence: intensive use of translation aids may foster a lasting reliance on the non-target language (e.g., L1). While this facilitates short-term access to meaning, it can reduce exposure to the target language and hinder the formation of autonomous representations (Cook, 2001; mixed findings on L1 glosses in Taylor, 2014). **STR response:** in Phase 1, support is integrated directly into the reading flow to avoid split attention; in Phase 2, support is removed entirely and meanings are actively retrieved, thereby systematically limiting long-term L1 reliance. **Caveat:** the risk of overusing Phase 1 remains and requires careful transition management. **Recommendation:** dose support context-sensitively and for limited time windows, reduce it stepwise,

schedule repeated readings without recourse to support languages, and assign tasks exclusively in the target language.

Moreover, reading alone conveys cultural and pragmatic dimensions only to a limited extent; complementary materials or communicative tasks are therefore advisable. Because language also transmits cultural and pragmatic practices, STR should be supplemented with materials that promote authenticity and intercultural sensitivity. Finally, although Phase 2 implements retrieval practice, it does so primarily within a narrow time window. Genuine spaced repetition arises only through repeated encounters with the same items in later chapters or texts.

To minimize these risks, we recommend consistent implementation of the second phase, careful selection of appropriate texts, and integration of supplementary pedagogical elements that more strongly address cultural, pragmatic, and productive dimensions of language learning.

Methodological Validation and Bias Risks

The L1-dependency index will be validated as a latent construct within an MTMM framework (behavior / self-report / performance); CFA will adhere to standard fit criteria, and adherence to fading will be logged protocol-wise. Planned studies follow current Open Science recommendations for preregistration in SLA research (Marsden, Mackey & Plonsky, 2019).

Construct validity: L1-dependency index and language autonomy.

The operationalizations proposed in Table 3 are testable but require dedicated validation studies. For the L1-dependency index (proportion of responses with explicit L1 recourse), we propose multimethod validation: (a) convergent validity via correlated behavioral markers (time spent on glosses / eye-tracking, frequency of L1 lookups, language choice in brief oral responses); (b) discriminant validity vis-à-vis general reading competence; (c) test–retest reliability across repeated sessions; (d) measurement invariance across proficiency levels and language pairs. Although planned as a longitudinal measure, the index should be modeled as a latent variable across multiple tasks/vignettes (MTMM) rather than as a simple item-level proportion.

Language autonomy should be modeled as a composite construct that combines (i) performance-sensitive indicators in Phase 2 (retrieval/comprehension without aids) with (ii) measures of L1 independence. To control for prior knowledge, we recommend: (1) an extensive pretest (VST/grammar/reading fluency) and subsequent stratified randomization; (2) item-response modeling to ensure level fairness and comparability of text items; (3) ANCOVA/linear mixed models with prior-knowledge covariates. The common heuristic “percentage of correct answers in Phase 2 vs. Phase 1” should be replaced by an **autonomy score**: an IRT-scaled Phase-2 performance metric, partially adjusted for Phase-1 performance and baseline competence (pretest), complemented by behavioral markers of L1 avoidance.

Manipulation and protocol fidelity (fading scaffolding).

To verify the core mechanism, a manipulation check is required: (a) a fading-integrity index (0/1 coding: “no aids in P2,” plus spot-check layout/protocol audits); (b) eye-tracking evidence of minimal gloss use in P1 and complete absence in P2 (fixation durations/regression rates); (c) process measures (keyboard/scroll logs) for adherence. This documents fading scaffolding quantitatively, not merely conceptually.

Quantifying the L1 crutch.

Beyond qualitative risk statements, we recommend predefined metrics: (i) session-specific **L1-reliance rate** (view/use time on L1 aids, share of L1 responses); (ii) trend/slope across sessions (expected negative slope under STR); (iii) thresholds for “successful fading” (e.g., < 5% L1 recourse in P2 by

week 4); (iv) survival analysis for “time to L1 independence.” These measures complement the planned indicators and render the risk assessment testable.

Risks in glossing (L1 over-reliance).

Meta-analytic and experimental findings indicate that glossing—especially L1-based and poorly designed—can induce shallow processing and persistent L1 use (Taylor, 2014; Boers et al., 2017; Boers, 2022). In STR, this risk is systematically constrained by minimally intrusive interlinear aids in Phase 1 and their categorical removal at the fading moment (Phase 2). We will test this empirically via the L1-reliance rate and its slope over sessions (Chapter 7).

Minimizing confounds (incl. prior knowledge).

In addition to pretest controls, we recommend: (a) measuring working memory/reading speed as covariates; (b) cluster or block randomization by class/level; (c) rotating parallel task forms to decouple practice and text-specific effects. Analyses should preferably use mixed models (random intercepts for persons and items).

Bias in RCTs: blinding and demand characteristics.

Full participant blinding is rarely feasible in pedagogical settings, but the following measures are practicable: (1) blinding of outcome coders (oral/paraphrased responses neutrally transcribed and coded by external raters); (2) automated, standardized tests (computer-based); (3) a preregistered analysis plan with explicit decision rules (ITT analysis, handling of missingness/attrition); (4) an active control condition with comparable novelty/load to dampen demand characteristics (neutral instructions, hypothesis masking); (5) quality/attention checks within motivation scales (e.g., embedded invert items). These measures complement the existing RCT framework.

Measurement plan and analysis (clarifying Table 3).

For convergent validation of language autonomy, we propose three task families: (A) inference questions targeting unknown but contextually inferable items (minimizing lexical prior knowledge), (B) brief L2 paraphrase tasks after Phase 2 (without L1 support), (C) dual-task retrieval (light secondary task) to test automated L2 access. All tasks will be IRT-scaled; reliability via McDonald's ω ; measurement-invariance tests across levels/language pairs. For the L1-dependency index, cross-indicator latent scores (eye-tracking, usage logs, language choice) will be modeled; its relation to Phase-2 performance will be preregistered (primary hypothesis: negative correlation). Table 3 remains an overview of instruments but will be explicitly labeled as provisional until validation metrics are available.

Validated instruments and ecological validity.

To calibrate the L1-dependency index and establish convergent validity for language autonomy, we will additionally collect validated questionnaires: LEAP-Q (Language Experience and Proficiency Questionnaire), BLP (Bilingual Language Profile), and BSWQ (Bilingual Switching Questionnaire). These scales will enter the models as covariates/moderators and serve to test convergent and discriminant validity of the newly introduced indices. As a theoretical frame for code-switching and usage profiles, we also reference Kroll & Dussias (2017). For ecological validity, fading scaffolding will be tracked longitudinally for at least eight weeks with weekly measurement points, enabling quantification of decreasing L1 dependence and the build-up of language autonomy as a trend. Motivational effects will be assessed using mixed methods: quantitatively (ANOVA/LMM on longitudinal data) and qualitatively (semi-structured interviews; learning diaries).

Summary.

The measures outlined in this section—construct validation (including LEAP-Q/BLP/BSWQ), manipulation checks for fading scaffolding, quantitative L1-dependency metrics, and bias mitigation in RCTs—increase the internal and external validity of findings. In doing so, the risks identified here are not merely acknowledged but rendered empirically measurable; Chapter 7 details their planned examination within the research program.

5.3 Applications of STR in Different Contexts

The potential applications of STR are diverse and encompass a wide range of teaching and learning environments.

In **self-directed learning**, the method is suitable not only for autonomous learners outside of formal courses but also for reactivating passive knowledge after extended breaks.

In **school and university language courses**, STR can serve as supplementary material for targeted enhancement of receptive competencies without overburdening learners with productive demands.

In **blended-learning formats**, Phase 2 offers particular potential: it can be digitally supported, repeated as often as desired, and allows for flexible, individualized learning paths as well as controlled withdrawal of support.

In **adult and senior education**, STR provides a low-threshold, low-stress entry through comprehensible texts, facilitating initial access to the target language and sustaining long-term motivation.

In **multilingual settings**, STR supports the immediate formation of target-language–meaning connections and reduces interference risks from non-target languages through gradual reduction of L1 support.

Didactic Potentials

- **Self-study.** Highly relevant for autonomous learners without institutional affiliation; equally useful for re-entry after learning breaks.
- **Group instruction.** The two-phase structure enables partner reading (role-based phase alternation) and collaborative activities (e.g., jigsaw tasks, peer questioning in Phase 2, brief micro-discussions about content).

Finally, the method can be **adapted internationally**, including for languages with divergent structures such as Turkish. In Phase 1, complex morphological phenomena (e.g., agglutination or relative constructions) can be made transparent through precise glosses before Phase 2 promotes autonomous meaning retrieval.

Example: Application of STR to Turkish

Phase 1 – with support:

Çocukluğumun en eski hatırası (the oldest memory of my childhood; *çocukluk* — *childhood*; *-um* — *possessive suffix, 1st person singular*; *en* — *superlative*; *eski* — *old, ancient*; *hatıra* — *memory*) bir ağustos şenliği gecesidir (a night of an August festival; *bir* — *one/a*; *ağustos* — *August*; *şenlik* — *festival, celebration*; *gece* — *night*; *-dir* — *copula, 3rd person singular*).

Bu hatıra (this memory; *bu* — *this*; *hatıra* — *memory*), görülmüş bir şeyden ziyade (rather than something seen; *görülmek* — *to be seen, passive of görmek* — *to see*; *bir şey* — *something*; *-den ziyade* — *more than, as compared to*) vaktiyle dinlenmiş bir masalın (of a once-heard fairy tale; *vaktiyle* — *once, formerly*; *dinlenmek* — *to be heard, passive of dinlemek* — *to listen*; *masal* — *fairy tale*; *-ın* — *possessive suffix, 3rd person singular*) hayalde bıraktığı izlere benzer (resembles the traces it left in the imagination; *hayal* — *imagination, dream*; *bırakmak* — *to leave, to let*; *-tığı* — *relative suffix, 3rd person singular*; *iz* — *trace*; *-ler* — *plural*; *benzemek* — *to resemble, to be similar*; *-r* — *present tense, 3rd person singular*).

Bu dünyada başka bir âlemde (in this world as if in another world; *dünya* — *world*; *başka* — *different, other*; *bir* — *one/a*; *âlem* — *world, realm, e.g., hayaller âlemi* — *world of dreams*) ucu bucağı olmayan bir bahçe (a boundless garden; *ucu bucağı olmayan* — *without visible boundaries*; *uç* — *edge, tip*; *bucak* — *corner*; *olmayan* — *non-existent*; *bahçe* — *garden*).

Phase 2 – without support:

Çocukluğumun en eski hatırası bir ağustos şenliği gecesidir.

Bu hatıra, görülmüş bir şeyden ziyade vaktiyle dinlenmiş bir masalın hayalde bıraktığı izlere benzer. Bu dünyada başka bir âlemde ucu bucağı olmayan bir bahçe.

Application note:

This example demonstrates how STR can clarify complex morphological structures of Turkish (e.g., agglutination, relative constructions like *-tığ*) and subtle semantic nuances (e.g., *ucu bucağı olmayan*) in Phase 1 through precise glosses to ensure comprehension. Phase 2 can then promote active retrieval and reduce dependence on L1.

AI-assisted tools can generate such glosses automatically by analyzing Turkish suffix combinations and idiomatic expressions, thereby facilitating the adaptation of authentic texts for learners at A1–B2 levels. Current overviews of chatbot-assisted learning show feasible didactic implementations and effectiveness factors compatible with STR workflows (Zhang, Zou & Cheng, 2023; see also Godwin-Jones, 2021).

We explicitly frame STR as a protocol, recognizing overlaps with assisted repeated reading and interlinear glossing, while emphasizing our contribution in the standardized binary coupling with predefined fading and measurable outcomes.

5.4 Framework of Novelty (Reframing)

STR stands within the tradition of practice-oriented two-phase formats (e.g., assisted repeated reading; interlinear glossing) but differs in presenting the **first academic operationalization** of a two-phase reading process with structured, binary fading. Phase 2 is **obligatory** (0% support, explicit retrieval requirement), and the **fading moment** is protocolled and verifiable.

This shifts the innovation claim from a mere *technique* (translation + repetition) toward a **verifiable protocol** with preregisterable outcomes and adherence/manipulation checks.

Distinction from Existing Approaches

- **Interlinear reading without fading:** Support is not removed; no retrieval requirement. STR introduces a clearly defined fading moment and a mandatory Phase 2.
- **Continuous/adaptive fading:** Support is reduced gradually or selectively; STR employs a categorical removal at a fixed point.
- **Repeated reading without support:** There is no initial comprehension assurance; STR combines comprehension assurance in P1 with active retrieval in P2.
- **Non-standardized two-phase formats (e.g., commercial interlinear + repetition models):** The second unsupported reading is optional and unmonitored. STR introduces mandatory fading with measurable indicators (e.g., L1-dependency index, delayed recall, load profiles) and defines adherence criteria for the binary P1→P2 transition. Moreover, these popularized models typically rely on unadapted original texts, whereas STR deliberately employs leveled A1–A2 materials for controlled research comparability.
- **Assisted repeated reading:** Combines support and repetition but does not systematically isolate the fading effect; in STR, this moment constitutes a **protocol-defined event** (preregistration, mechanism testing; relief in P1 vs. targeted activation in P2).

Core Novelty Elements of STR

- **Protocol and adherence control:** Fixed parity of time/exposure, preregistered analysis plan, adherence/manipulation checks (layout audit, process and eye-tracking data) for the fading moment.
- **Measurable constructs:** L1-dependency index (with longitudinal dynamics), designated *fading-moment* marker, autonomy score; analyses via longitudinal and mixed models.

- **Standardized outcomes:** Delayed recall, phase-specific cognitive load profiles, and *time to L1 independence* (survival analysis).

Consequence

STR systematizes previously scattered practices within a coherent theoretical–methodological framework (*IR*→*ER*, transitional extensive reading) and establishes the prerequisites for **replicability and comparability** in future studies.

5.5 Hypotheses for Future Research

The central innovation of STR lies in its **systematic two-phase structure**: Phase 1 facilitates immediate comprehension of the text, while Phase 2 requires **autonomous meaning retrieval**. This dual structure distinguishes STR from existing reading methods and forms the foundation for the following hypotheses.

1. **Vocabulary acquisition.**

STR may lead to greater retention of target vocabulary among beginners (A1–A2) compared to other reading methods (e.g., marginal glossing, parallel texts) in both immediate and delayed post-tests. This effect would stem directly from the two-phase architecture — the combination of comprehension assurance (Phase 1) and obligatory retrieval (Phase 2).

2. **Text comprehension.**

Learners at intermediate levels (B1–B2) may achieve higher global and local reading comprehension with STR than control groups using other approaches.

3. **Reduction of linguistic dependence.**

Through its second phase, STR could reduce the risk of persistent translation dependence (reliance on support languages such as L1). Empirically, this could be tested through comparative studies (e.g., STR vs. interlinear reading) that measure the **direction of meaning retrieval** — exclusively in the target language versus target → support language(s) — and quantify their relative frequency.

4. **Variation by language combination.**

The effectiveness of STR may depend on the **typological distance** between L1 and the target language, as well as on the integration of additional *spaced-repetition* techniques. Effects may vary across language pairs — for instance, being stronger where morphological complexity or typological distance is high.

5. **Motivation.**

STR may enhance **reading motivation** more effectively than traditional glossing or parallel-text approaches, as learners experience immediate success (“reading and understanding everything”) in Phase 1 while perceiving growing autonomy in Phase 2. This structure combines short-term assurance with long-term competence development. The motivational advantage aligns with the **Self-Determination Theory** (Ryan & Deci, 2020), which identifies competence and autonomy experiences as central drivers of intrinsic motivation.

6. Conclusion and Outlook

6.1 Summary

Supported Two-Phase Reading (STR) integrates immediate comprehension assurance with obligatory meaning retrieval in a clearly defined procedural model. It is positioned as a **bridge approach between intensive and extensive reading** (*transitional extensive reading*), utilizing narrative long-form texts and thus maintaining compatibility with extensive reading principles (see Section 2.2).

Phase 1 reduces cognitive load, while Phase 2 consolidates lexical meaning through active retrieval (see Section 3).

Outlook: Priority now lies with empirical validation of effectiveness, operationalization of the *fading* process, and implementation across formal, informal, and AI-assisted learning environments.

STR should not be understood as a “new method,” but as a **refined and measurable protocol**. The *fading moment* functions as a clearly defined target and evaluation point that systematically integrates repeated reading, comprehension assurance, and retrieval, making them comparable across studies.

In perspective, STR procedures can be integrated into **AI-assisted environments** (e.g., adaptive support density, automated manipulation checks), as suggested by recent reviews of chatbot-based learning scenarios (Zhang, Zou & Cheng, 2023).

6.2 Limitations (aligned with Section 3.4)

- **Focus on receptive skills:** Productive competences (speaking, writing) benefit only indirectly.
 - **Material suitability:** Effectiveness depends on text quality and level-appropriateness; overly complex texts increase cognitive load and reduce motivation.
 - **Dependence risks:** Risk of L1/L2 crutch when aids are overused; Phase 2 must be implemented consistently.
 - **Repetition logic:** Phase 2 occurs immediately; *spaced-repetition* effects arise only through repeated encounters in later sections. It also remains unclear to what extent STR effects persist over longer time spans (months or years); long-term studies are not yet available.
 - **Cultural/pragmatic scope:** Reading-based formats alone capture cultural and pragmatic aspects only to a limited extent.
-

6.3 Research Perspectives (condensed; without duplication of Chapter 7)

- **Effectiveness:** Randomized comparative studies of STR versus alternative approaches using identical materials and extended intervention periods.
 - **Mechanisms:** Eye-tracking, pupillometry, and EEG designs to differentiate cognitive relief (P1) from retrieval activation (P2).
 - **Autonomy:** Longitudinal designs tracking the *L1-dependency index* and motivational measures, including transfer and delayed-recall testing.
 - **Note:** Detailed study designs and variables are outlined in Chapter 7.
-

6.4 Concluding Remarks

STR addresses key bottlenecks of traditional reading approaches by combining **accessible comprehension (without a dictionary)** with the **step toward autonomy** within a single structured framework.

If future studies confirm the expected effects, the approach holds promise for both classroom instruction and self-directed learning: it can reduce disruptions to reading flow, strengthen contextual vocabulary retention, and stabilize learner motivation (see Sections 2.2, 3, and 5.1).

Thus, STR establishes a **practical bridge between interlinear reading and extensive reading**, systematically integrating **retrieval practice** and **categorical fading** (a binary, section-bound transition without residual aids). This provides clear perspectives for didactic applications and further empirical research.

The effectiveness claims formulated here remain **tentative and hypothesis-driven**; their verification will be the focus of the studies outlined in **Chapter 7**.

7. Research Program

To systematically examine the effectiveness of **Supported Two-Phase Reading (STR)**, we propose a multi-stage research program encompassing exploratory pilot studies, controlled laboratory experiments, and extended classroom interventions.

The objective is not only to determine the pedagogical relevance of STR but also to empirically capture its **cognitive effect profile** and underlying mechanisms.

7.1 Pilot Study (Exploratory–Didactic)

- **Participants:** Approximately 40 learners (A1–A2 level, school and adult education), randomly assigned to two groups.
- **Duration:** 8 weeks, with 2–3 reading sessions per week (20–30 minutes each).
- **Conditions:**
 - Experimental group: STR (Phase 1 + Phase 2)
 - Control group: Marginal glossing
 - Optional (if sample size allows): Interlinear reading without Phase 2 → direct comparison “scaffold with fading” vs. “scaffold without fading.”
- **Materials:** Narrative texts containing ~60 target words (new but high-frequency), partially supplemented with audio.

Operationalization (Measurement Instruments)

- **Vocabulary tests:** Immediate and delayed recall, including transfer tasks.
- **Autonomous meaning retrieval:** Number of correctly recalled target words in retrieval tasks without aids.
- **L1-dependency index:** Proportion of responses showing explicit reliance on L1 (translations, L1 anchors, language choice). Lower values indicate greater linguistic autonomy.
- **Cognitive load:**
 - Subjective: NASA-TLX questionnaire.
 - Objective: Reading time per 100 words; optionally, eye-tracking (fixation duration, regression rates) and pupillometry (load index).
- **Motivation:** Likert scales, standardized questionnaires (e.g., LLOS), qualitative interviews, and learner diaries.
- **Linguistic autonomy:** Proportion of correctly answered comprehension questions in Phase 2 vs. Phase 1; self-assessments.

Validated Scales (Baseline & Follow-up)

In addition to domain-specific performance tests, the following validated instruments will be administered at **T0 (baseline)** and **T2 (post-test)**:

LEAP-Q (language experience/proficiency profile), BLP (language dominance/usage), and BSWQ (switching tendencies).

These scales serve as covariates in ANCOVA/LMM analyses and as convergent validation tools for the L1-dependency index (e.g., fixation time on L1 aids, language choice in responses).

For theoretical framing of interference and language mode, we additionally refer to **Grosjean (2010)** and **Kroll & Dussias (2017)**.

Longitudinal Tracking of Fading Scaffolding

The intervention spans ≥ 8 weeks with weekly measurements of **L1 reliance rate** and **autonomy score** (T0–T8).

Primary parameters:

- (i) slope of L1 reliance rate (expected negative trend under STR),
- (ii) time to L1 independence (survival analysis),
- (iii) stability of the autonomy score.

Mixed-Methods Design for Motivation

- **Quantitative:** ANOVA/LMM (Group $\times$ Time) on standardized motivation scales.
- **Qualitative:** Semi-structured interviews (T2) and learner diaries (sample subset), thematically coded and triangulated with quantitative data.

Procedure Note: LEAP-Q / BLP / BSWQ – Scoring and Analysis

- **Administration:** At T0 (baseline) and T2 (post-test).
- **Scoring & standardization:** Subscales according to manuals; raw scores z-standardized within sample (optionally stratified by L1/location).
- **Outliers:** Winsorized at 1st/99th percentile or excluded at $|z| > 3$ (pre-registered).
- **Missing data:**
 - Item level: $\leq 10\%$ missing $\rightarrow$ person-mean imputation within subscale; 10–20% $\rightarrow$ subscale treated as missing.
 - Person level: $> 20\%$ missing across scales $\rightarrow$ case exclusion (sensitivity analysis with/without exclusion); additionally, multiple imputation ($m=20$) under MAR/MCAR; main analyses ITT with robust SE.
- **Model integration:** z-scores as covariates in ANCOVA/LMM; exploratory moderation (Scale $\times$ Group $\times$ Time).
- **Composite index (optional):** “Bilingual Profile (z-composite)” = mean of z-subscales; report reliability (ω); if $\omega < .70$, report subscales separately.
- **Pre-registration:** All rules for standardization, outlier handling, and missing data must be fixed before data inspection.



Table 3. Operationalization of Key Constructs

Construct	Definition	Measurement Instruments / Indicators
Vocabulary growth	Number of correctly recalled target words after reading	Immediate & delayed recall tests; transfer tasks in new contexts
Autonomous meaning retrieval	Retrieval of target words without support	Proportion of correctly recalled words in Phase 2; retrieval tasks without aids
L1-dependency index	Proportion of responses showing explicit L1 reliance	Language choice in translation tasks; annotations in learner diaries; analysis of oral responses

Construct	Definition	Measurement Instruments / Indicators
Cognitive load	Mental effort during reading	NASA-TLX; reading time per 100 words; eye-tracking (fixation duration, regressions); pupillometry
Motivation	Enjoyment, autonomy, willingness to read	LLOS; Likert scales; interviews; learner diaries
Linguistic autonomy	Ability to comprehend texts without L1 aids	Proportion of correctly answered comprehension questions in Phase 2 vs. Phase 1; self-assessments

Analysis

- ANOVA (Group $\times$ Time)
- Effect sizes (Cohen's d , η^2)
- Linear mixed models (random intercepts for items and participants) for robust inference
- Qualitative content analysis of interviews and learner diaries

7.2 Controlled Laboratory Studies (Mechanistic Level)

Objective: To capture the cognitive mechanisms of STR in detail.

- **Eye-tracking (cf. Warren et al., 2018):** Fixation durations and regression rates as indicators of comprehension processes; longer fixations and more regressions signal increased cognitive load.
- **Pupillometry:** Pupil dilation as a load index — an objective measure of cognitive effort, allowing fine-grained temporal tracking of mental load.
- **EEG/fNIRS (cf. Shen et al., 2021):** Theta/alpha-band activations and increased blood flow in frontal areas during meaning retrieval — evidence for retrieval practice and deep processing.
- **Subjective ratings:** Supplementary self-assessments of perceived effort.
- **Comparison Phase 1 vs. Phase 2:** Immediate differences in working memory load, processing depth, and cognitive activation.

7.3 Classroom Interventions (RCT Designs)

- **Sample:** $N > 100$; duration ≥ 8 –10 weeks.
- **Comparison groups:**
 - STR
 - Marginal glossing
 - Parallel texts
 - Interlinear reading without Phase 2
 - Additional condition: reading without any support (control against circular inference; cf. Choi & Zhang, 2022).

Measures

- **Vocabulary growth:** Immediate and delayed recall tests; transfer tasks in new contexts; retention intervals (1–4 weeks).
- **Global & local comprehension:** Multiple-choice and short-answer tasks for detail understanding.

- **Reading fluency:** Words per minute (speed) and accuracy.
- **Motivation:** LLOS; Likert scales on reading enjoyment and effort; qualitative interviews; learner diaries.
- **Linguistic autonomy:** Proportion of correctly solved tasks without L1 support; self-assessments.
- **L1-dependency index:** As defined in 7.1, tracked across sessions as an indicator of decreasing L1 reliance.

Objective: Demonstrate long-term effectiveness and motivational impact under real classroom conditions, and isolate the effects of the **two-phase structure** (scaffolding + fading).

7.4 Summary of the Research Program

The research program integrates **mechanistic studies** (eye-tracking, pupillometry, EEG/fNIRS) with **didactic interventions** (pilot and RCT studies), enabling evaluation of:

- **Whether STR works** (evidence of effectiveness)
- **Why it works** (mechanistic explanation)

A key contribution lies in the **empirical isolation of fading scaffolding** through:

- **Difference in meaning retrieval:** Proportion of correctly recalled target words in Phase 2 vs. Phase 1 (immediate and delayed recall tests).
- **Cognitive relief:** Shorter fixation durations and lower regression rates in Phase 1; greater pupil dilation and longer processing times in Phase 2 — indicators of targeted cognitive activation.
- **Autonomy indicator:** Reduction of L1 usage across sessions, measurable via the L1-dependency index.
- **Motivation:** Stabilization or increase in reading enjoyment (Likert scales) after repeated exposure to the two-phase structure.

Together, these studies open a wide research field in which STR can be empirically validated and theoretically refined as both a **didactically innovative procedure** and a **bridge framework between intensive and extensive reading** (*IR→ER; transitional extensive reading*).

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Appendix: Practical Experiences with Supported Two-Phase Reading (STR) – Anecdotal / Non-Representative

The following statements are based on a qualitative analysis of over **300 publicly available Amazon reviews** of bilingual English–German books designed according to the principles of **Supported Two-Phase Reading (STR)**. In the main text, quotations are used solely for illustrative purposes.

Ethics and Data Protection

No interventions involving human participants were conducted. The analyzed materials were drawn from open platforms (online reviews) and from voluntarily submitted feedback forms; their use complies with the terms and conditions of the respective platforms.

All quotations and data were fully anonymized: potentially identifying metadata (names/usernames, locations, exact timestamps, product IDs) were removed or aggregated to the month/year level. The analysis is **anecdotal and non-representative**, serving exclusively for **hypothesis generation**.

Since this conceptual contribution contains no original experimental data, **formal ethical approval was not required**. For future empirical studies (e.g., RCTs, eye-tracking experiments), preregistration and ethical clearance will be obtained where applicable.

Appendix A: Qualitative Analysis of Feedback (Sampling, Coding, Anonymization)

- **Categorization:** Thematic content analysis; quotations selected illustratively.
 - **Coding:** Conducted by one researcher; interrater reliability not yet established. For future studies, two independent raters are planned.
 - **Exclusions:** Pure star ratings without text and clear duplicates were excluded; longer quotations were minimally shortened or anonymized (without altering meaning).
 - **Quotation selection & de-identification:** Direct quotes were screened for identifying details and time-aggregated (month/year).
 - **Limitations:** Self-selection and platform bias are possible; findings are non-representative and serve for hypothesis generation, not for evidence of effect.
 - **Data/material availability:** Aggregated data, codebook, and analysis scripts will be made available on an **Open Science platform** after peer review; raw texts of the quotes will not be published.
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Table A1. Thematic Categories and Exemplary Quotations

Category	Exemplary Quotations	Observation
Immediate text comprehension without dictionary	“Finally an English book where I don’t need a dictionary.” (Wolfgang S., 2023) “Understanding becomes much easier thanks to the structure and repetition.” (Bastienne, 14 July 2024)	Direct access to texts; noticeable reduction in cognitive load
Vocabulary acquisition in context	“I start learning the words automatically.” (A. Fischer, 9 April 2021) “Over time, you just know what each word means.” (Hasibutzi, 25 July 2020)	Incremental vocabulary learning without isolated memorization
Motivation and enjoyment	“I couldn’t stop reading because I wanted to know how the story ends.” (Marie, 21 April 2020) “Thanks to the repetition, the words are easy to remember.” (Tinamunk, 10 May 2021)	Continuous reading flow enhances motivation and enjoyment
Structure and two-phase format	“No more annoying dictionary lookups.” (FrauLyna, 17 Dec 2024) “First vocabulary and expressions, then the same text only in English.” (Roland K., 26 May 2023)	Positive perception: short sections and direct transition from Phase 1 → Phase 2
Broad target-group applicability	“I’m over 50 [...] It’s really fun, and you learn with joy.” (Anja, 2022) “I gave it to my 75-year-old mother – she gets along perfectly.” (k., 24 Feb 2025)	Suitable for diverse age groups, including older learners
Grammar intuition and implicit learning	“I recognize verbs in different forms and understand the meaning without translation.” (Elena K., 2021) “You learn that automatically while reading.” (J.H., 2022)	Implicit acquisition of grammatical structures through repeated exposure
Emotional engagement and storytelling	“The story is fascinating – you want to keep reading and learn at the same time.” (Sven, 4 May 2020)	Narrative interest and emotional engagement reinforce learning effects

Category	Exemplary Quotations	Observation
Critiques and improvement potential	“Unfortunately, the German translation is not always very good.” (Klaus, 20 May 2021) “The left page is too close to the binding.” (brombeis, 20 Apr 2024)	Criticism of translation quality, layout, and topic variety — led to subsequent improvements (editing, layout, e-books)

Summary Observation

The practical experiences largely **confirm the core theoretical assumptions** of STR: immediate comprehension, contextual vocabulary development, motivational enhancement, implicit grammar acquisition, and flexible applicability across learner groups. At the same time, critical feedback provides valuable insights for **further optimization** of translation quality, design, and thematic diversity.